Flavor annotation: pre-run review

prompt design, target molecules, and expected yield — for sign-off before the run

```
WHAT IS BEING PROPOSED

Annotate 163,120 molecules from coconut-flavordb that carry no measured
flavor label, using an LLM, and keep ONLY the labels the model attributes to
literature recall. Those become the sparse flavor half of MolPallete's two-part
condition vector; the pocket half stays zero during pretraining and is filled
from tastepocket during finetuning.

Model: deepseek/deepseek-v4-flash        Estimated cost: ~$16
Vocabulary: 24 constrained labels        Mode: one API call per molecule

WHY ONLY THE `documented` TIER

A condition vector must carry information the encoder cannot already derive
from its own input. Anything computed FROM the molecular graph fails that test:
np_classifier class, RDKit fragment bits, and LLM labels inferred from
functional groups are all f(structure). Conditioning a structure model on
f(structure) is the leak that made the original 97-bit condvec unusable --
removing it moved rank1_distinct from 680 to 5,851.

Literature recall is different in kind. "Amarogentin is a benchmark bitterant"
is a fact about the world, not about the graph, and the encoder has no way to
reach it. That is the only tier worth keeping.

Note this reverses the ranking from the earlier accuracy benchmark, where a
deterministic class prior beat the LLM on total correct labels (78% vs 53%).
It won BECAUSE it is f(structure) -- which makes it worthless here. Accuracy and
leakiness are different properties; that benchmark measured the wrong one.

WHAT THIS DOCUMENT COVERS

  1. the exact prompt, and the measurement behind each rule
  2. the 163,120 target molecules
  3. worked examples with real model output
  4. model selection evidence
  5. expected yield and the decision it implies
```

1. The prompt, and why each rule is there

docs/flavor_annotation_prompt.md — reproduced verbatim in section 1b

RULE	MOTIVATED BY (measured)
Constrained 24-label vocabulary	FlavorDB's raw set is 715 descriptors with 253 singletons and 67% of mass on sweet/sweet-like. The folded 24 cover 94.3% of genuine assignments; free text would invite invention across a 715-way space.
Explicit `unknown`, declared "expected and correct"	Most natural products have never been tasted. Without a licence to abstain the model confabulates on exactly the compounds we are asking about. Measured: 83% abstention on unmeasured compounds when permitted.
`odorless` separated from `unknown`	`odorless` asserts non-volatility; `unknown` asserts ignorance. Collapsing them would contaminate the ~209K physically-derived negative class with abstentions.
Evidence tier required (documented / close_analog / structural / none)	Makes confidence falsifiable. `documented` must name a source, so fabricated provenance is checkable. This is the field the whole pipeline filters on.
"`structural` alone is usually insufficient", with the numbers quoted at the model	Class rules are weak and were measured: alkaloid->bitter is 19% precise; glycosides split between glycyrrhizin (sweet) and amarogentin (bitter). Stating the numbers in the prompt suppresses the class shortcut -- measured 0/5 bitter on alkaloid-tagged compounds across every model tested.
Organism inference banned	Source organism does not predict flavor: mango's compounds are 4% "fruity" against a 2% base; Capsicum is DEPLETED in bitter.
Confidence tied to tier	A second, independent calibration signal.
`glycoside_heavy` warning in the user block	"sugar-bearing => sweet" is right ~6% of the time once the imputed sweet-like class is excluded. This is the trap two benchmark compounds are built to spring.

1b. Prompt — verbatim

this exact text is sent as the system message

```
You are a flavor chemist annotating compounds for a research dataset that will
train a molecular model. Your annotations become training labels, so a confident
wrong answer is far more damaging than an admission of ignorance. Most compounds
you are shown are natural products that have **never been sensorially
characterised by anyone** — for those, the correct answer is `unknown`.

### Allowed labels (use ONLY these)

**Taste:** sweet, bitter, sour, salty, umami
**Odor:** fruity, green, floral, fatty, woody, spicy, roasted, sulfurous,
earthy, nutty, herbal, medicinal, citrus, dairy, alcoholic, meaty, minty
**Special:** odorless, unknown

`odorless` and `unknown` are different claims and must not be substituted for
each other:
- `odorless` = a positive claim that the compound has no perceptible odor
  (typically non-volatile: high molecular weight, highly polar, involatile salt).
  A compound can be `odorless` and still taste sweet or bitter — if so, give both.
- `unknown` = you do not know. Use it alone, with no other label.

### Evidence tier (required, one of)

- **`documented`** — you recall this *specific compound's* reported sensory
  profile. You must name the source in `source`: a flavor/fragrance database
  (The Good Scents Company, Flavornet, FlavorDB, BitterDB, SuperSweet, Pyrfume),
  a regulatory/industry listing (FEMA GRAS, JECFA, IOFI), or a specific
  publication. If you cannot name where you know it from, this is not
  `documented`.
- **`close_analog`** — you do not recall this compound, but you recall a closely
  related one whose profile is very likely to carry over. Name that compound in
  `analog` and say why the property should transfer.
- **`structural`** — inference from functional groups or compound class alone
  (e.g. "it is a glycoside", "it is an ester", "it is an alkaloid").
- **`none`** — no basis. Then `labels` must be exactly `["unknown"]`.

### Rules

1. At most 3 labels, ordered by confidence. Fewer is better.
2. **`structural` evidence alone is usually insufficient.** Class-level rules are
   weak: alkaloids are bitter only ~19% of the time, and glycosides split between
   intensely sweet (glycyrrhizin) and intensely bitter (amarogentin). If your only
   basis is the compound class, prefer `["unknown"]` unless the inference is
   near-certain (e.g. a simple sugar; an involatile inorganic salt).
3. Do not infer flavor from the source organism. A plant contains hundreds of
   compounds and only a few carry its taste.
4. `confidence` in [0,1] must reflect the evidence tier. `documented` with a
   named source may exceed 0.8; `structural` should rarely exceed 0.4.
5. Never invent a source, a publication or an analog. If you are unsure whether
   a source really documents this compound, use `close_analog` or `structural`.
6. Annotate every input id exactly once.

### Output
```

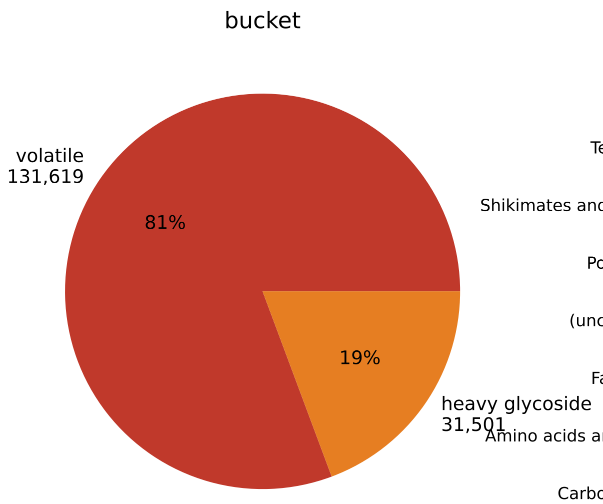
1b. Prompt — verbatim (continued)

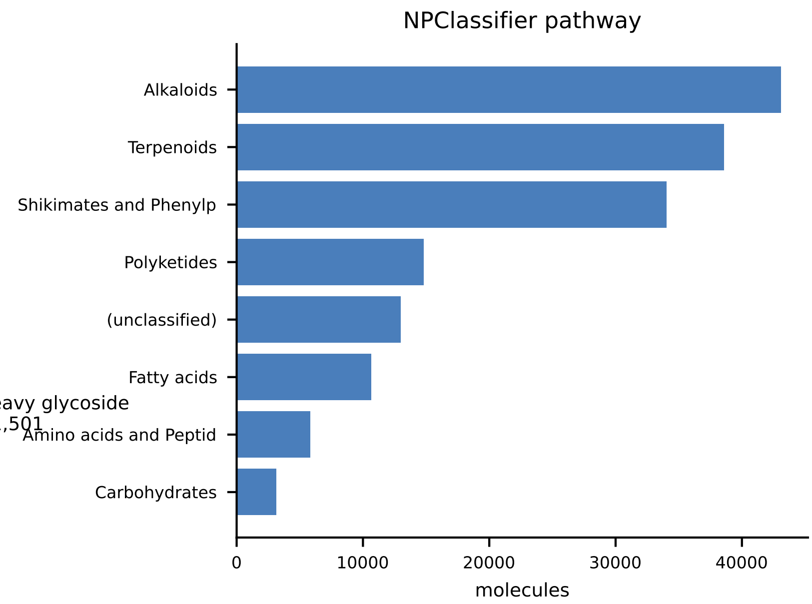
JSON only, no prose:

````
```json
{"results":[
  {"id":"...",
   "labels":["..."],
   "confidence":0.0,
   "evidence":"documented|close_analog|structural|none",
   "source":"…name it, or null",
   "analog":"…compound name, or null",
   "reasoning":"one sentence, max 25 words"}
]}
```
````

2. The target molecules

163,120 molecules with no measured flavor label






```
  bucket
    volatile_unmeasured    131,619   MW <= 350; could be odorant OR tastant
    glycoside_heavy         31,501   MW > 350 + sugar; cannot smell, CAN taste

  molecular weight        median 305   p5 166   p95 608

  naming
    any name                85,095   52.2%
    real chemical name      78,409   48.1%
    catalogue id only        6,686   (SCHEMBL/ZINC/CID/CAS)
    NO name at all          78,025   47.8%
```

THE NAMING NUMBER IS THE ONE THAT MATTERS

Just under half these molecules have a real chemical name; the rest are bare CNP identifiers or catalogue codes. A compound with no name has almost certainly never been sensorially characterised, so the correct output for it is `unknown` -- and that is why abstention discipline, not accuracy, was the gating criterion in model selection.

It also sets the expectation for yield. The models score 43-68% `documented` on FlavorDB compounds precisely because those are characterised. These are not.

COMPOSITION IS NOT FLAVOR CHEMISTRY

Alkaloids 26.4%, terpenoids 23.7%, shikimates 20.9%. This is a natural-products library -- plant secondary metabolites, many non-volatile, many never tasted, some toxic. It is not a flavor-and-fragrance catalogue, and treating it as one is the error that would produce 163,120 confident inventions.

3. Worked examples — real model output

20 benchmark compounds; ground truth from FlavorDB, sweet-like excluded

ROUND 1 — recognisable compounds (Claude Opus 5, same prompt)

compound	predicted	truth	tier	hit
Glycyrrhizic acid	sweet, odorless	odorless,spicy, sweet	documented	YES
Amarogentin	bitter, odorless	bitter	documented	YES
Neoquassin	bitter, odorless	bitter	close_analog	YES
Swertiamarin	bitter, odorless	bitter	close_analog	YES
Isoquercetin	bitter, odorless	odorless	close_analog	YES
phenacetin	odorless, bitter	bitter	documented	YES
4-methyl-2-pentanone	fruity	dairy,fruity,green	documented	YES
allyl nonanoate	fruity, fatty	alcoholic,dairy, fatty,fruity	documented	YES
N,N-dimethylthioformamide	sulfurous	bitter	structural	no
alkylpyrazine (CAS only)	roasted,nutty, earthy	fruity	close_analog	no

THE PAIR THAT MATTERS Amarogentin (MW 586 glycoside, among the most bitter substances known) and glycyrrhizic acid (MW 823 glycoside, genuinely sweet). Both correct, both `documented` with named sources. A model applying "glycoside => sweet" gets one of them wrong; this is the circularity test and the prompt passes it.

Both misses are defensible and were the two lowest-confidence answers in the set. A volatile thioamide plausibly IS sulfurous (FlavorDB's lone "bitter" looks incomplete); alkylpyrazines are textbook roasted/nutty and FlavorDB calling one "fruit" is the more suspect label.

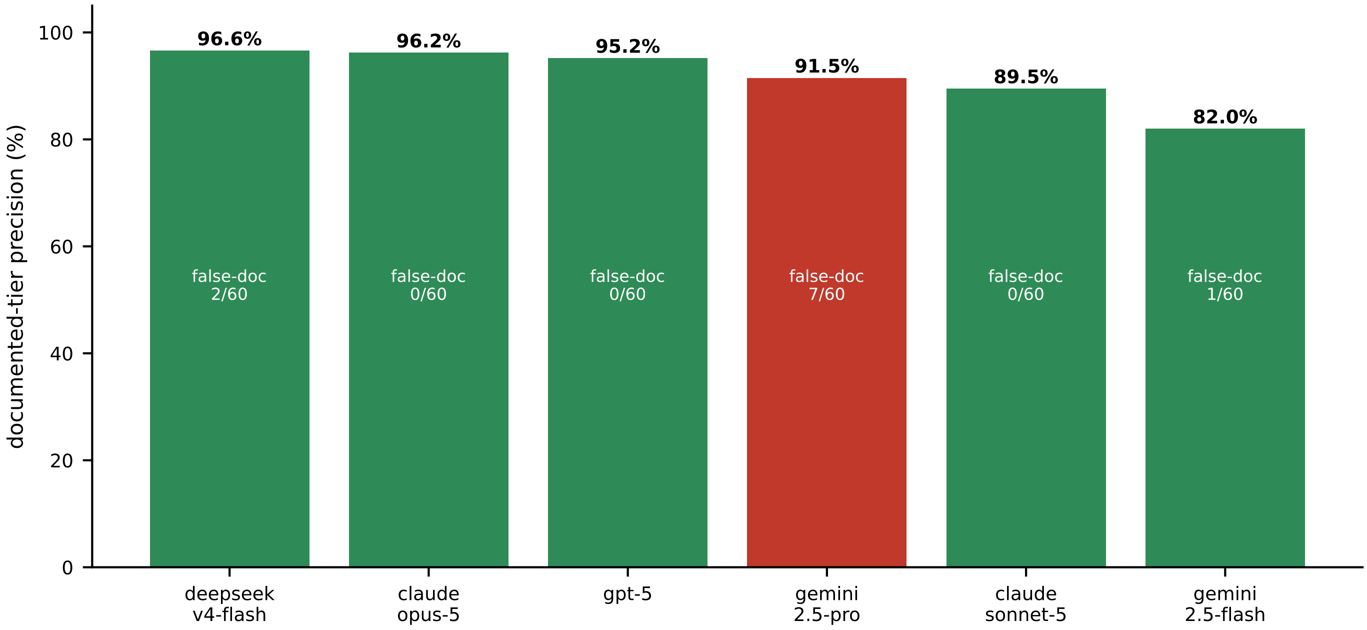
ROUND 2 — obscure compounds, 7 of 10 with NO measured label

6 of 7 returned ["unknown"], evidence "none", confidence 0.0.
The 7th returned `odorless` on a permanently charged quaternary benzylisoquinolinium, arguing a fixed cation cannot volatilise -- which is the same physics used to derive the 209K negative class, reached independently.
0 of 5 alkaloid-tagged compounds were labelled bitter.
It also flagged that NPClassifier had mislabelled a synthetic Biginelli scaffold as an alkaloid.

That is the behaviour the run depends on, on molecules resembling the real targets.

4. Model selection

180 compounds each (120 measured + 60 unmeasured), identical prompt



model	doc yield	doc precision	false-doc	no-resp	163K
deepseek-v4-flash	48.3%	96.6% [88,99]	2/60	5	$16
claude-opus-5	43.3%	96.2% [87,99]	0/60	0	$1,219
gpt-5	17.5%	95.2% [77,99]	0/60	26	$384
gemini-2.5-pro	68.3%	91.5% [83,96]	7/60	36	$384
claude-sonnet-5	31.7%	89.5% [76,96]	0/60	0	$487
gemini-2.5-flash	50.8%	82.0% [71,90]	1/60	0	$94

gemini-2.5-pro is DISQUALIFIED despite the best yield. It named a literature source for 7 of 60 compounds nobody has characterised, and failed to respond on 30% of the set. Fabricated provenance voids the premise of keeping this tier.

deepseek-v4-flash is selected. Its documented precision is statistically tied with Opus 5 (overlapping CIs), yield is comparable, and false-documented rates are indistinguishable at n=60 -- at 1/76 the cost.

Reasoning models were worse AND dearer on this task, which wants a short structured verdict rather than deliberation: qwen3.8-max burned 8,000 output tokens on 10 compounds and produced nothing; deepseek-v4-pro reasoned itself INTO the alkaloid class-prior trap that cheaper models avoided.

5. Expected yield, and the decision it implies

estimated from the 60-compound unmeasured arm; 500-molecule run in progress

MEASURED YIELD

```
  (500-molecule run still in flight at time of writing)

  From the 60-compound unmeasured arm of the model benchmark:
    deepseek-v4-flash   documented 2/60   abstained 52/60
    claude-opus-5       documented 0/60   abstained 47/60
    gpt-5               documented 0/60   abstained 50/60

  Extrapolated to 163,120 targets: roughly 0 - 5,000 documented labels.
```

WHY IT IS LOW, AND WHY THAT IS THE HONEST ANSWER

The models return 43-68% `documented` on FlavorDB compounds and near zero on the targets. That is not a failure -- it is the same wall every other route hit. These are obscure plant secondary metabolites; 51.9% have no real chemical name at all. There is no literature to recall because nobody has tasted them.

WHAT THE RUN BUYS

```
  measured labels already in corpus        21,387     5.44%
  + documented LLM labels (expected)      ~1,600 - 5,000
  ------------------------------------------------------
  flavor coverage after the run             ~5.8% - 6.7%
```

A gain of roughly 0.4 - 1.3 percentage points on a 393,066-molecule corpus.

RECOMMENDATION

Run it -- but on the arithmetic, not on optimism. At ~$16 it is close to free, every label is literature-recalled rather than structure-derived, and the pipeline is already benchmarked. What it will NOT do is make flavor a usable pretraining condition: 6.7% coverage has the same problem 5.4% did, which is that an all-zero vector for 93% of molecules teaches source provenance rather than flavor.

The value is in the finetuning stage, where the flavor bits sit alongside pocket embeddings on a much smaller, deliberately selected set.

BEFORE ACCEPTING ANY OF IT

1. SHUFFLE TEST. Permute flavor condvecs across instances and re-score retrieval against the H@1 0.3726 baseline. Unchanged means the bits are dead weight; a collapse beyond what permutation justifies means they leak.
2. Keep `documented` ONLY. `close_analog` is partly inferred from structure and does not have the exogeneity that makes this defensible.
3. Record `label_source` per molecule so measured, LLM-documented and absent never mix silently.
4. Distinguish ZERO from ABSENT in the vector. An all-zero condvec must mean "unknown", not "no flavor", or the model learns that the modal molecule is flavourless.